

SYSTEM OF CIRCLES



SYNOPSIS

Angle between two intersecting circles:

- If d is distance between the centres, r_1, r_2 are the radii of two intersecting circles then the angle

between the circles is $\cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1r_2} \right)$.

- If $d^2 = r_1^2 + r_2^2$ then the angle between the two circles is 90° . In this case the length of their

common chord is $\frac{2r_1r_2}{\sqrt{r_1^2 + r_2^2}}$.

- If θ is the angle between two circles with radii r_1 and r_2 then length of the common chord is

$$\frac{2r_1r_2 \sin \theta}{\sqrt{r_1^2 + r_2^2 + 2r_1r_2 \cos \theta}}$$

- If θ is angle between the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and

$$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0 \text{ then}$$

$$\cos \theta = \frac{(c + c_1) - 2(gg_1 + ff_1)}{2\sqrt{g^2 + f^2 - c} \sqrt{g_1^2 + f_1^2 - c_1}}.$$

- If the circles cut each other orthogonally then $2gg_1 + 2ff_1 = c + c_1$.

- Two circles with centres c_1, c_2 and radius " r " cut

each other orthogonally. Then $r = \frac{c_1c_2}{\sqrt{2}}$.

- Two circles of radii r_1, r_2 cut orthogonally then area included between the circles is

$$r_1^2 \tan^{-1} (r_2 / r_1) + r_2^2 \tan^{-1} (r_1 / r_2) - r_1r_2.$$

- If P and Q are conjugate points with respect to the circle $S = 0$ then the circle on PQ as diameter cuts the circle $S = 0$ orthogonally.

Radical Axis :

- The locus of a point which moves such that its powers with respect to two circles are equal, is a line called radical axis.

- The radical axis of two circles is a straight line perpendicular to the line joining the centres of the two circles.

- The equation of the radical axis of the circles $S = 0$ and $S^1 = 0$ is $S - S^1 = 0$

- If two circles intersect each other then their common chord is radical axis of the two circles.

- If two circles touch each other then their radical axis is the common tangent at the point of contact.

- The locus of the centre of a circle which cuts the given two circles orthogonally is the radical axis of the given two circles.

- If one circle lies entirely in the other then the radical axis lies outside of both the circles.

- If one circle lies between outside the other then radical axis lies in between both the circles.

- The number of radical axes of n circles is nC_2 when no three of their centres are collinear.

- There is no radical axis for concentric circles.

Radical Centre :

- The point of intersection of the radical axes of three circles taken in pairs whose centres are not collinear is called the radical centre of the three circles.

- The lengths of the tangents from the radical centre of 3 circles to the 3 circles are equal.

- If P is the radical centre of three circles and PA is the length of the tangent from P to one of the three circles then the circle whose centre is P and radius is PA cuts the three circles orthogonally.

- The radical centre of the three circles described on the sides of a triangle as diameters is the orthocentre of the triangle.

- If A, B, C are the centres of three circles which cut each other orthogonally then the radical centre of the three circles is the orthocentre of the triangle ABC.

- If A, B, C are the centres of three circles which touch each other externally then the radical centre of the 3 circles is the in-centre of the triangle ABC.



LEVEL-I-(C.W)

ANGLE BETWEEN TWO INTERSECTING CIRCLES

- If radii of two circles are 4 and 3 and distance between centres is $\sqrt{37}$ then angle between the circles is
1) 30° 2) 45° 3) 60° 4) 90°
- The angle between the circles $x^2 + y^2 - 4x - 6y - 3 = 0$, $x^2 + y^2 + 8x - 4y + 11 = 0$,
1) $\pi/2$ 2) $\pi/4$ 3) $\pi/3$ 4) $\pi/6$
- The angle at which the circles $x^2 + y^2 + 8x - 2y - 9 = 0$, $x^2 + y^2 - 2x + 8y - 7 = 0$ intersect is
1) obtuse 2) $\pi/6$
3) $\pi/3$ 4) $\pi/2$
- The angle between the circles $x^2 + y^2 = a^2$, $x^2 + y^2 = ax + ay$ is
1) $\frac{\pi}{4}$ 2) $\frac{3\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3}$

RADICAL AXIS OF TWO CIRCLES

- The locus of the centre of a circle which cuts the circles $2x^2 + 2y^2 - x - 7 = 0$ and $4x^2 + 4y^2 - 3x - y = 0$ orthogonally is a straight line whose slope is
1) -1 2) 1 3) -2 4) $-\frac{5}{2}$
- Observe the following statements:
I. The lengths of the tangents from any point on the line $2x + 3y = 5$ to the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 4x + 6y = 19$ are equal in length.
II. There is only one point such that the tangents from it to the three given non-concentric circles whose centres non-collinear are equal in length. Then the correct statement is:
(1) Only I (2) Only II
(3) Both I & II (4) Neither I nor II
- If the circles $x^2 + y^2 - 10x + 2y + 10 = 0$ and $x^2 + y^2 - 4x - 6y - 12 = 0$ touch each other then the slope of the common tangent at the point of contact of the circles is
1) $3/4$ 2) $4/3$ 3) $-4/3$ 4) $-3/4$

- The perpendicular distance from the origin to the radical axis of the circles $2x^2 + 2y^2 - 3x - y + 3 = 0$ and $3x^2 + 3y^2 - x + y - 1 = 0$ is
1) $\sqrt{2}$ 2) $\frac{11}{\sqrt{74}}$ 3) $\frac{\sqrt{5}}{2}$ 4) $\sqrt{\frac{5}{2}}$

- The slope of the radical axis of the circles $(x + 2)^2 + (y + 3)^2 = 25$ and $(x + 1)^2 + (y - 1)^2 = 25$ is
1) $-1/4$ 2) $1/4$ 3) -4 4) $-1/2$
- The equation of the radical axis of the two circles $7x^2 + 7y^2 - 7x + 14y + 18 = 0$ and $4x^2 + 4y^2 - 7x + 8y + 20 = 0$ is given by
(1) $3x^2 + 3y^2 - 6y - 2 = 0$
(2) $21x - 68 = 0$
(3) $x - 2y - 5 = 0$
(4) $x + 2y + 5 = 0$

RADICAL CENTRE AND ORTHOGONAL CIRCLES

- Radical centre of $x^2 + y^2 - x + 3y - 3 = 0$, $x^2 + y^2 - 2x + 2y + 2 = 0$ and $x^2 + y^2 + 2x + 3y - 9 = 0$ is
1) (2, 3) 2) (3, 2) 3) (-2, 3) 4) (-3, -2)
- The number of points such that the tangents from it to three given circles are equal in length, is
1) 1 2) 2 3) 3 4) 4
- The radical axis of two touching circles divides the line segment joining the centre of circles in the ratio of their
1) areas 2) radii 3) 1:1 4) 1:-1
- If the radical centre of the three circles $x^2 + y^2 - 2x - 1 = 0$, $x^2 + y^2 - 3y = 1$ and $2x^2 + 2y^2 - x - 7y - 2 = 0$ is Q then $Q_x + Q_y =$
1) 3 2) 0 3) 1 4) -1
- A: The radical centre of the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 3x = 4$, $x^2 + y^2 - 4y = 4$ is (0, 0)
R: Radical centre of three circles is the point of concurrence of the radical axes of the circles taken in pairs.
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not the correct explanation of A
3) A is true but R is false
4) A is false but R is true

16. Circles with radical centre as centre and radius equals to length of tangent from radical centre to any of the three circles will

- 1) Bisects the circumference of all the three circles
- 2) Bisects the circumference of at least one of the circle
- 3) Orthogonal to all the three circles
- 4) Orthogonal to at least one of the circle

17. The radical centre of the circles $(x - 1)^2 + (y - 2)^2 = 341$, $(x - 4)^2 + (y - 1)^2 = 341$, $(x - 5)^2 + (y - 4)^2 = 341$ is

- 1) (3, 3) 2) (4, 1) 3) (6, 6) 4) $\left(\frac{10}{3}, \frac{7}{3}\right)$

18. The radius of one circle is twice the radius of another circle whose centres are (2, 0), (1, 2) respectively cutting orthogonally. Then the radius of the first circle is

- 1) 1 2) 2 3) 3 4) 5

19. The circle with centre (2, 3) and intersecting $x^2 + y^2 - 4x + 2y - 7 = 0$ orthogonally has the radius

- 1) 1 2) 2 3) 3 4) 4

20. The circle $2x^2 + 2y^2 + px + 6y - 10 = 0$ and $3x^2 + 3y^2 + 15x + py + 21 = 0$ are orthogonal then p =

- 1) 7/8 2) 5/8 3) 8/7 4) 8/5

21. If the circles $x^2 + y^2 + 2a^1x + 2b^1y + c^1 = 0$ and $2x^2 + 2y^2 + 2ax + 2by + c = 0$ intersect orthogonally, then

- 1) $aa^1 + bb^1 = c + c^1$ 2) $aa^1 + bb^1 = c + \frac{c^1}{2}$

- 3) $aa^1 + bb^1 = \frac{c}{2} + c^1$ 4) $2(aa^1 + bb^1) = c + c^1$

22. Two circles of equal radii 'r' cut orthogonally. If their centres are (-2, -3) and (-5, -6), then r =

- 1) 1 2) 2 3) 3 4) 4

23. The equation of the circle passing through (0, 0) and cutting the circles $x^2 + y^2 + 6x - 15 = 0$, $x^2 + y^2 - 8y + 10 = 0$ orthogonally is

- 1) $(x + 5/2)^2 + (y - 5/4)^2 = \frac{125}{16}$

2) $x^2 + y^2 - 5x - 5y = 0$

3) $2(x^2 + y^2) - 10x - 5y = 0$

4) $x^2 + y^2 - 5x + 5y = 0$

24. If the circles $(x + a)^2 + (y + b)^2 = a^2$, $(x + \alpha)^2 + (y + \beta)^2 = \beta^2$ cut orthogonally then $\alpha^2 + b^2 =$

- 1) $a\alpha + b\beta$ 2) $a^2 + \beta^2$
- 3) $-2(a\alpha + b\beta)$ 4) $2(a\alpha + b\beta)$

25. Radical axis exists for

- 1) any two circles
- 2) any two concentric circles
- 3) any two non-concentric circles
- 4) Can't say

COMMON CHORD OF THE CIRCLES

26. Two circles whose radii are r and R and whose distance between the centres is 'd' cut each other orthogonally. Then the length of their common chord is

- 1) $\frac{2rR}{\sqrt{r^2 + R^2}}$ 2) $\frac{rR}{\sqrt{r^2 + R^2}}$

- 3) $\frac{2rR}{\sqrt{r^2 + R^2}}$ 4) $\frac{2rR}{r^2 + R^2}$

27. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ bisects the circumference of the circle $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ then the length of the common chord of the circles is

- 1) $2\sqrt{g_1^2 + f_1^2 - c_1}$ 2) $\sqrt{g_1^2 + f_1^2 - c_1}$

- 3) $\sqrt{g^2 + f^2 - c}$ 4) $2\sqrt{g^2 + f^2 - c}$

28. If 3, 4 are the radii and 5 is the distance between the centres of two intersecting circles then the length of the common chord of the circles is

- 1) 12/5 2) 24/25
- 3) 24/5 4) 5/24

29. If the length of the common chord of two circles $x^2 + y^2 + 8x + 1 = 0$ and

$x^2 + y^2 + 2\mu y - 1 = 0$ is $2\sqrt{6}$ then $\mu =$

- 1) ± 2 2) ± 4
- 3) ± 8 4) ± 3

30. The distance of the point (1, 2) from the common chord of the circles $x^2 + y^2 + 6x - 16 = 0$ and $x^2 + y^2 - 2x - 6y - 6 = 0$ is

- 1) 1 2) 1/5
- 3) 5 4) 2

31. If the circle $x^2+y^2+2gx+2fy+c=0$ bisects the circumference of the circle $x^2+y^2+2g_1x+2f_1y+c_1=0$ then

- 1) $2g_1(g-g_1)+2f_1(f-f_1)=c-c_1$
- 2) $2g_1(g-g_1)+2f_1(f-f_1)+c-c_1=0$
- 3) $g_1(g-g_1)+f_1(f-f_1)=c-c_1$
- 4) $2g(g-g_1)+2f_1(f-f_1)=c-c_1$

32. If the circle $3x^2+3y^2+10x+y-27=0$ bisects the circumference of the circle $x^2+y^2=k$ then $k^2-1=$

- 1) 27 2) 728 3) 9 4) 80

LEVEL-I (C.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 3 | 2) 3 | 3) 4 | 4) 2 | 5) 1 | 6) 3 |
| 7) 1 | 8) 2 | 9) 1 | 10) 2 | 11) 1 | 12) 1 |
| 13) 2 | 14) 2 | 15) 1 | 16) 3 | 17) 1 | 18) 2 |
| 19) 2 | 20) 3 | 21) 3 | 22) 3 | 23) 3 | 24) 4 |
| 25) 3 | 26) 3 | 27) 1 | 28) 3 | 29) 4 | 30) 1 |
| 31) 1 | 32) 4 | | | | |

LEVEL-I (C.W.) - HINTS

1. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1r_2} \right)$
2. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1r_2} \right)$
3. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1r_2} \right)$
4. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1r_2} \right)$
5. Slope of R.A
6. Definition of R.A
7. slope of R.A
8. Find distance from (0,0) to R.A
9. Slope of $S-S'=0$
10. $S-S'=0$
11. Intersection of R.A of any two pairs of given circles
12. Def. of Radical centre
13. Draw the diagram for touching circles
14. Intersection of R.A. of any two pairs of circle
15. Intersection of R.A. of any two pairs of circle

16. Def. of orthogonal circle
17. Intersection of R.A. of any two pairs of circle
18. $c_1c_2=d$, and radii are $2r_1, r_2$ and use $d^2 = r_1^2 + r_2^2$
19. $2gg^1 + 2ff^1 = c + c^1$
20. $2gg^1 + 2ff^1 = c + c^1$
21. $2gg^1 + 2ff^1 = C + C^1$
22. use $d^2 = r_1^2 + r_2^2$, $d = C_1C_2$
23. Let $S=0$ be the req circle and use orthogonality condition
24. $2gg^1 + 2ff^1 = c + c^1$
25. Definition of radical axis
26. Put $\theta = 90^\circ$ in $\frac{2rR \sin \theta}{\sqrt{r^2 + R^2 + 2rR \cos \theta}}$
27. Diameter of later circle
28. $\theta = 90^\circ \Rightarrow L.C.C = \frac{2r_1r_2}{\sqrt{r_1^2 + r_2^2}}$
29. $2\sqrt{r^2 - d^2} = 2\sqrt{6}$
30. Distance from (1,2) to R.A
31. Common chord of two circles passes through $(-g_1, -f_1)$
32. R.A. passes through (0,0)



LEVEL-I(H.W)

ANGLE BETWEEN TWO INTERSECTING CIRCLES

1. If radii are $2, \sqrt{2}$ and distance between centres is $\sqrt{2}$ then the angle between the circle is
 - 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{3\pi}{4}$
2. The angle between the tangents from a point on $x^2 + y^2 + 2x + 4y - 31 = 0$ to the circle $x^2 + y^2 + 2x + 4y - 4 = 0$ is
 - 1) $\pi/6$ 2) $\pi/2$ 3) $\pi/4$ 4) $\pi/3$
3. The angle between the circles $x^2 + y^2 - 12x - 6y + 41 = 0$, $x^2 + y^2 + kx + 6y - 59 = 0$ is $\frac{\pi}{4}$ then $k =$
 - 1) ± 2 2) ± 1 3) ± 4 4) ± 3

4. The angle between the circles

$$x^2 + y^2 - 2lx + g = 0, x^2 + y^2 + 2my - g = 0 \text{ is}$$

- 1) $\frac{\pi}{4}$ 2) $\frac{3\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3}$

RADICAL AXIS OF TWO CIRCLES

5. The locus of the centre of the circles which intersects the circles $x^2 + y^2 = 1$ and

$$x^2 + y^2 - 2x + y = 0 \text{ orthogonally is}$$

- 1) a line whose equation is $2x - y - 1 = 0$
2) a line whose equation is $2x + y = 1$
3) a circle 4) a pair of lines

6. The equations of two circles are

$$x^2 + y^2 + 2\lambda x + 5 = 0 \text{ and } x^2 + y^2 + 2\lambda y + 5 = 0.$$

P is any point on the line $x - y = 0$. If PA and PB are the lengths of the tangents from P to the two circles and $PA = 3$ then $PB =$

- 1) 1.6 2) 6 3) 4 4) 3

7. If the locus of the centre of the circle which cuts the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and

$$x^2 + y^2 - 4x + 6y + 4 = 0 \text{ orthogonally is}$$

$ax + by + c = 0$, then the ascending order of a, b, c is

- 1) a, b, c 2) b, c, a 3) c, a, b 4) a, c, b

8. The distance of (1, -2) from the common chord of $x^2 + y^2 - 5x + 4y - 2 = 0$ and $x^2 + y^2 - 2x + 8y + 3 = 0$ is

- 1) 2 2) 1 3) 0 4) 3

9. The slope of the radical axis of the circles $x^2 + y^2 + 3x + 4y - 5 = 0$ and $x^2 + y^2 - 5x + 5y - 6 = 0$ is

- 1) 1 2) 3 3) 5 4) 8

10. The radical axis of circles $x^2 + y^2 + 3x + 4y - 5 = 0$ and $x^2 + y^2 - 5x + 5y - 6 = 0$ is

- 1) $8x + y + 1 = 0$ 2) $8x - y + 1 = 0$
3) $8x - 8y + 1 = 0$ 4) $-8x + y + 1 = 0$

RADICAL CENTRE AND ORTHOGONAL CIRCLES

11. Radical centre of the circles

$$x^2 + y^2 + 4x - 7 = 0,$$

$$2x^2 + 2y^2 + 3x + 5y - 9 = 0, x^2 + y^2 + y = 0 \text{ is}$$

- 1) (2, 1) 2) $\left(\frac{7}{3}, \frac{11}{3}\right)$ 3) (7, 11) 4) $\left(\frac{7}{2}, \frac{11}{2}\right)$

12. If $S_1 = 0, S_2 = 0$ and $S_3 = 0$ are the three circles whose radical centre is the point P, then the lengths l_1, l_2, l_3 of the tangents from P to the three circles are such that

- 1) $l_1 = 2l_2 = 3l_3$ 2) $l_1 = l_2 = l_3$
3) $l_1 \neq l_2 \neq l_3$ 4) $l_1 = l_2 \neq l_3$

13. The radical axis of the circles $x^2 + y^2 + 4x - 6y = 12$ and $x^2 + y^2 + 2x - 2y - 1 = 0$ divides the line segment joining the centres of the circles in the ratio

- 1) 27 : 17 2) 3 : 7 3) -27 : 17 4) -3 : 7

14. If Q is the radical centre of the three circles $x^2 + y^2 = a^2, (x - g)^2 + y^2 = a^2$ and $x^2 + (y - f)^2 = a^2$ then $Q_x + Q_y =$

- 1) $g + f$ 2) $\frac{g + f}{2}$ 3) $2g + 2f$ 4) $\frac{-g - f}{2}$

15. Match the following :

Circles Radical centre

I. $x^2 + y^2 = 1, x^2 + y^2 - 2x = 1,$ a) (0, 0)

$$x^2 + y^2 - 2y = 1$$

II. $x^2 + y^2 - x + 3y - 3 = 0,$ b) (2, 3)

$$x^2 + y^2 - 2x + 2y + 2 = 0$$

$$x^2 + y^2 + 2x + 3y - 9 = 0$$

III. $x^2 + y^2 - 8x + 40 = 0,$ c) (8, -15/2)

$$x^2 + y^2 - 5x + 16 = 0$$

$$x^2 + y^2 - 8x + 16y + 160 = 0$$

- 1) a, b, c 2) b, c, a 3) c, a, b 4) a, c, b

16. The point from which the lengths of tangents to the three circles $x^2 + y^2 - 4 = 0, x^2 + y^2 - 2x + 3y = 0$ and $x^2 + y^2 + 7y - 18 = 0$ are equal is

- 1) (2, 5) 2) (3, 4) 3) (4, 3) 4) (5, 2)

17. The equation of the circle which cuts the three circles $x^2 + y^2 = a^2, (x - g)^2 + y^2 = a^2$ and $x^2 + (y - f)^2 = a^2$ orthogonally is

- 1) $x^2 + y^2 - 2gx - 2fy + a^2 = 0$
2) $x^2 + y^2 - gx - fy + a^2 = 0$
3) $x^2 + y^2 - fx - gy + a^2 = 0$
4) $x^2 + y^2 + gx + fy - a^2 = 0$

18. If the circles of equal radius and centres at (2, 3), (5, 6) cut orthogonally, then the radius of one of the circles is

- 1) 3 2) $3\sqrt{2}$ 3) 6 4) 4

19. The circle cutting the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ orthogonally and having centre $(-1, 2)$ is

- 1) $x^2 + y^2 + 2x - 4y - 2 = 0$
- 2) $x^2 + y^2 + 2x - 4y + 2 = 0$
- 3) $x^2 + y^2 - 2x + 4y - 2 = 0$
- 4) $x^2 + y^2 + 2x - 4y - 4 = 0$

20. If the circle $x^2 + y^2 + 2x - 2y + 4 = 0$ cuts the circle $x^2 + y^2 + 4x + 2fy + 2 = 0$ orthogonally, then $f =$

- 1) 1 2) 2 3) -1 4) -2

21. A: If $x^2 + y^2 - 2x + 3y + k = 0$,

$x^2 + y^2 + 8x - 6y - 7 = 0$, cut each other orthogonally then $k = 10$

R: The circles $x^2 + y^2 + 2gx + 2fy + c = 0$,

$x^2 + y^2 + 2g'x + 2f'y + c' = 0$ cut each other orthogonally if $2gg' + 2ff' = c + c'$.

- 1) Both A and R are true and R is the correct explanation of A
- 2) Both A and R are true but R is not the correct explanation of A
- 3) A is true but R is false
- 4) A is false but R is true

22. If the circles of same radius 'a' and centres at $(2, -3)$, $(5, 6)$ cut orthogonally then $a =$

- 1) 4 2) $4\sqrt{2}$ 3) $3\sqrt{5}$ 4) 3

23. I: The equation of the circle cutting orthogonally the circles

$$x^2 + y^2 - 8x - 2y + 16 = 0,$$

$x^2 + y^2 - 4x - 4y - 1 = 0$ and passing through the point $(1, 1)$ is

$$3x^2 + 3y^2 - 14x + 23y - 15 = 0.$$

II: The equation of the circle which cuts orthogonally the three circles

$$x^2 + y^2 + 2x + 17y + 4 = 0,$$

$$x^2 + y^2 + 7x + 6y + 11 = 0,$$

$$x^2 + y^2 - x + 22y + 3 = 0$$

$$x^2 + y^2 - 6x - 4y - 44 = 0 \text{ is}$$

- 1) Only I is true 2) Only II is true
- 3) both I & II are true 4) neither I nor II true

24. Match the following:

I: If $x^2 + y^2 - 6x - 8y + 12 = 0$, a) 1

$$x^2 + y^2 - 4x + 6y + k = 0$$

cut orthogonally then $k =$

II: If $x^2 + y^2 - 2x + 3y + k = 0$, b) -10

$$x^2 + y^2 + 8x - 6y - 7 = 0$$

cut orthogonally then $k =$

III: If $x^2 + y^2 + 2x - 2y + 4 = 0$, c) -24

$$x^2 + y^2 + 4x - 2ky + 2 = 0$$

cut orthogonally then $k =$

- 1) a, b, c 2) b, c, a 3) c, b, a 4) a, c, b

25. Radical centre of the circles

$$x^2 + y^2 + 4x - 6y - 12 = 0,$$

$$x^2 + y^2 + 4x - 6y + 2 = 0,$$

$$x^2 + y^2 + 4x - 6y + 4 = 0$$

- 1) $(0, 0)$ 2) $(-2, 3)$ 3) $(2, -3)$ 4) does not exist

26. Two circles of radii r and R intersect at an acute angle θ . The length of their common chord is

- 1) $\frac{2rR \sin \theta}{\sqrt{r^2 + R^2 - 2rR \cos \theta}}$ 2) $\frac{2rR \sin \theta}{\sqrt{r^2 + R^2}}$
- 3) $\frac{2rR \sin \theta}{\sqrt{R^2 - r^2}}$ 4) $\frac{2rR \sin \theta}{\sqrt{r^2 + R^2 + 2rR \cos \theta}}$

27. If the circle $x^2 + y^2 - 2x - 2y - 1 = 0$ bisects the circumference of the circle $x^2 + y^2 = 1$ then the length of the common chord of the circles is

- 1) 1 2) 2 3) $\sqrt{3}$ 4) $2\sqrt{3}$

28. The length of the common chord of the two circles of radii 10, 24 whose centres are 26 units apart is

- 1) $\frac{10}{13}$ 2) 240 3) $\frac{240}{13}$ 4) $\frac{120}{13}$

29. The length of the common chord of

$$x^2 + y^2 + 2x + 3y + 1 = 0,$$

$$x^2 + y^2 + 4x + 3y + 2 = 0 \text{ is}$$

- 1) $\sqrt{2}$ 2) 2 3) $2\sqrt{2}$ 4) $6\sqrt{2}$

30. The distance of $(1, -2)$ from the common chord of $x^2 + y^2 - 5x + 4y - 2 = 0$ and $x^2 + y^2 - 2x + 8y + 3 = 0$ is

- 1) 2 2) 1 3) 0 4) 3

31. If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle

$$x^2 + y^2 + 2x - 6y - 15 = 0 \text{ then } k =$$

- (1) 21 (2) -21 (3) 23 (4) -23

32. If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle

$$x^2 + y^2 - 2x + 8y - d = 0 \text{ then } c + d =$$

- 1) 60 (2) 50 (3) 40 (4) 30

33. The equation of the circle whose diameter is the common chord of the circles

$$x^2 + y^2 + 2x + 3y + 2 = 0$$

$$x^2 + y^2 + 2x - 3y - 4 = 0 \text{ is}$$

1) $x^2 + y^2 + 2x + 2y + 2 = 0$

2) $x^2 + y^2 + 2x + 2y - 1 = 0$

3) $x^2 + y^2 + 2x + 2y + 1 = 0$

4) $x^2 + y^2 + 2x + 2y + 3 = 0$

LEVEL-I (H.W.) - KEY

- 1) 4 2) 4 3) 3 4) 3 5) 1 6) 4 7) 2
8) 3 9) 4 10) 2 11) 1 12) 2 13) 3 14) 2
15) 1 16) 4 17) 2 18) 1 19) 1 20) 3 21) 4
22) 3 23) 3 24) 3 25) 4 26) 4 27) 2 28) 3
29) 3 30) 3 31) 4 32) 2 33) 3

LEVEL-I (H.W.) - HINTS

1. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1 r_2} \right)$ 2. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1 r_2} \right)$

3. $\cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1 r_2} \right) = \frac{\pi}{4}$

4. $\theta = \cos^{-1} \left(\frac{d^2 - r_1^2 - r_2^2}{2r_1 r_2} \right)$

7. Compare R.A. with $ax + by + c = 0$

8. Distance from $(1, -2)$ to $S - S' = 0$

9. slope of $S - S' = 0$ 10. $S - S' = 0$

11. Intersection of R.A. of any two pairs of given circles

12. Def. of Radical centre 13. $-L_{11} / L_{22}$

14 to 16. Intersection of R.A. of any two pairs of circle

17. Find radical centre and its radius is length of tangent from it to any one of given circles

18. Use $d^2 = r_1^2 + r_2^2$, $d = C_1 C_2$

20, 21. $2gg^1 + 2ff^1 = c + c^1$

22. Use $d^2 = r_1^2 + r_2^2$, $d = C_1 C_2$

24. use orthogonality condition

25. For concentric circles R.A. is not defined.

27. Diameter of later circle

28. $\theta = 90^\circ \Rightarrow L.C.C = \frac{2r_1 r_2}{\sqrt{r_1^2 + r_2^2}}$

29. $2\sqrt{r^2 - d^2}$

30. Distance from $(1, -2)$ to R.A.

31. R.A. passes through centre of later circle

32. R.A. passes through $(1, -4)$

33. Substituting the centre of $S + \lambda L = 0$ in the equation of common chord of the given circles.



LEVEL-II-(C.W)

ANGLE BETWEEN TWO INTERSECTING CIRCLES

1. If the angle between the two equal circles with centres $(-2, 0), (2, 3)$ is 120° then the radius of the circle is

- 1) 5 2) 3 3) 1 4) 2

RADICAL AXIS, RADICAL CENTRE AND ORTHOGONAL CIRCLES

2. The locus of a point such that difference of the squares of the tangents from it to two given circles is constant, is given by

- 1) A circle
2) A line perpendicular to radical axis
3) A line parallel to radical axis
4) A pair of straight lines

3. The locus of the centre of a circle which cuts orthogonally the circle $x^2 + y^2 - 20x + 4 = 0$ and touches the line $x = 2$ is

- 1) $y^2 = 16x + 4$ 2) $x^2 = 16y + 4$
3) $x^2 = 16y$ 4) $y^2 = 16x$

4. If a circle passes through $(1, 2)$ and cuts $x^2 + y^2 = 4$ orthogonally then the locus of its centre is

- 1) $2x + 4y - 9 = 0$ 2) $x + y - 3 = 0$
3) $x + y - 9 = 0$ 4) $2x + 3y = 7$

5. A circle passes through origin and has its centre on $x = y$ cuts $x^2 + y^2 - 4x - 6y + 10 = 0$ orthogonally, then its equation is

- 1) $x^2 + y^2 - x - y = 0$ 2) $x^2 + y^2 - 4x - 4y = 0$
3) $x^2 + y^2 - 2x - 2y = 0$ 4) $x^2 + y^2 - 3x - 3y = 0$

6. The circle through $(-2, 5)$, $(0, 0)$ and intersecting the circle $x^2 + y^2 - 4x + 3y - 1 = 0$ orthogonally is
- $2x^2 + 2y^2 - 11x - 16y = 0$
 - $x^2 + y^2 - 4x - 4y = 0$
 - $x^2 + y^2 + 2x - 5y = 0$
 - $x^2 + y^2 + 2x - 5y + 1 = 0$
7. $x = 1$ is the radical axis of two circles which cut each other orthogonally. If $x^2 + y^2 = 9$ is the equation of one circle then the equation of the other circle is
- $x^2 + y^2 - 9x + 9 = 0$
 - $x^2 + y^2 + 18x - 9 = 0$
 - $x^2 + y^2 - 18x + 9 = 0$
 - $x^2 + y^2 + 9x + 9 = 0$
8. In n ($n \geq 3$) circles the centres of no three circles are collinear. If the number of the radical axes of the circles is equal to the number of the radical centres of the circles then $n^2 - 4n - 5 =$
- 5
 - 0
 - 50
 - 7
9. The radical centre of three circles described on the sides of a triangle as diameters is of the $\triangle ABC$
- Orthocentre
 - Centroid
 - in-centre
 - Circumcentre
10. If the radical centre of $x^2 + y^2 - 4x + 2y + 3 = 0$, $x^2 + y^2 - x + 4y + 4 = 0$ and $x^2 + y^2 + 2gx + 5y + 7 = 0$ is $(-1, 1)$ then $g =$
- 3
 - 3
 - $-3/2$
 - $3/2$

COMMON CHORD OF THE CIRCLES

11. The line $2x + 3y = 1$ intersects the circle $x^2 + y^2 = 4$ at A and B. If the equation of the circle on AB as diameter is $x^2 + y^2 + 2gx + 2fy + c = 0$ then $c =$
- 50
 - $-54/13$
 - $50/13$
 - $-50/13$
12. If the line $x \cos \alpha + y \sin \alpha = p$ and the circle $x^2 + y^2 = a^2$ intersect at A and B then the equation of the circle on AB as diameter is $(x^2 + y^2 - a^2) + k(x \cos \alpha + y \sin \alpha - p) = 0$ then $k =$
- p
 - $-p$
 - $-4p$
 - $-2p$
13. The equation of the circle passing through the origin and the points of intersection of the circles $x^2 + y^2 - 4x - 6y - 3 = 0$, $x^2 + y^2 + 4x - 2y - 4 = 0$
- $x^2 + y^2 + 28x + 18y = 0$
 - $x^2 + y^2 - 18x - 28y = 0$
 - $x^2 + y^2 - 28x + 18y = 0$
 - $x^2 + y^2 - 28x - 18y = 0$
14. The equation of the circle describes on the common chord of the circles $x^2 + y^2 + 2x = 0$ and $x^2 + y^2 + 2y = 0$ as diameter is
- $x^2 + y^2 - x - y = 0$
 - $x^2 + y^2 + x - y = 0$
 - $x^2 + y^2 + x + y = 0$
 - $x^2 + y^2 - x + y = 0$
15. If $a \neq b$ then the length of common chord of the circles $(x-a)^2 + (y-b)^2 = c^2$ and $(x-b)^2 + (y-a)^2 = c^2$ is
- $\sqrt{4c^2 - 2(a-b)^2}$
 - $\sqrt{c^2 - (a-b)^2}$
 - $\sqrt{3c^2 - (a-b)^2}$
 - $\sqrt{2c^2 - (a-b)^2}$
16. The locus of the centre of the circle which cuts the circle $x^2 + y^2 - 20x + 4 = 0$ orthogonally and touches the line $x = 2$ is
- $y^2 = 4x$
 - $y^2 = 16x$
 - $x^2 = 4y$
 - $x^2 = 16y$
17. $x^2 + y^2 = a^2$ and $(x-c)^2 + y^2 = b^2$ are two intersecting circles. If a, b, c are the sides BC, CA, AB of $\triangle ABC$. If p_1, p_2, p_3 are the altitudes through A, B, C respectively then the length of the common chord is
- $2p_1$
 - $2p_2$
 - $2p_3$
 - p_1
18. The radius of the least circle passing through the point $(8, 4)$ and cutting the circle $x^2 + y^2 = 40$ orthogonally is
- $\sqrt{5}$
 - $\sqrt{7}$
 - $2\sqrt{5}$
 - $4\sqrt{5}$
19. (a, c) and (b, c) are the centres of two circles whose radical axis is the y-axis. If the radius of first circle is r then the diameter of the other circle is
- $2\sqrt{r^2 - b^2 + a^2}$
 - $\sqrt{r^2 - a^2 + b^2}$
 - $(r^2 - b^2 + a^2)$
 - $2\sqrt{r^2 - a^2 + b^2}$
20. B and C are points on the circle $x^2 + y^2 = a^2$, A point $A(b, c)$ lies on that circle such that $AB = AC = d$. The equation to BC is
- $bx + ay = a^2 - d^2$
 - $bx + ay = d^2 - a^2$
 - $bx + cy = 2a^2 - d^2$
 - $2(bx + cy) = 2a^2 - d^2$
21. If the circle $x^2 + y^2 + 8x - 4y + c = 0$ touches the circle $x^2 + y^2 + 2x + 4y - 11 = 0$ externally and cuts the circle $x^2 + y^2 - 6x + 8y + k = 0$ orthogonally then $k =$
- 59
 - 59
 - 19
 - 19

LEVEL-II (C.W.) - KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) 1 | 2) 3 | 3) 4 | 4) 1 | 5) 3 | 6) 1 | 7) 3 |
| 8) 2 | 9) 1 | 10) 4 | 11) 4 | 12) 4 | 13) 4 | 14) 3 |
| 15) 1 | 16) 2 | 17) 3 | 18) 1 | 19) 4 | 20) 4 | 21) 2 |

LEVEL-II (C.W.) - HINTS

- $\cos \theta = \frac{d^2 - r_1^2 - r_2^2}{2r_1r_2}$ where $r_1 = r_2$ and $\theta = 120^\circ$
- $S_{11} - S_{11}^1 = \lambda \Rightarrow$ locus of p is $S - S^1 = \lambda$
 $\Rightarrow$ radical axis + constant = 0 $\Rightarrow$ parallel to R.A
- Let $S = 0$ be req. circle and use given conditions
- Let the centre of the circle be (x_1, y_1)
- ${}^nC_2 = {}^nC_3 \Rightarrow n = 5$
- Intersection of R.A. of any two pairs of circle
- $S + \lambda L = 0$ has centre on $L = 0$
- $S + \lambda L = 0$ passes through $(0,0)$
- $S + \lambda L = 0$ has centre on $L = 0$
- Apply $2\sqrt{r^2 - d^2}$
- Let the circle $S = x^2 + y^2 + 2gx + 2fy + c = 0$ and apply orthogonal condition
- Draw the diagram
- Let the circle $S = x^2 + y^2 + 2gx + 2fy + c = 0$ and apply orthogonal condition
- In the radical axis put $x = 0$
- Consider the circle centre at (b, c) and radius d
- $C_1C_2 = r_1 + r_2$



LEVEL-II-(H.W)

ANGLE BETWEEN TWO INTERSECTING CIRCLES

- If the angle between two equal circles with centres $(3,10), (-5,4)$ is 120° then the radius of the circle is
1) 10 2) 5 3) 2 4) 1

RADICAL AXIS, RADICAL CENTRE AND ORTHOGONAL CIRCLES

- From any point P tangents of length t_1 and t_2 are drawn to two circles with centre A, B and if PN is the perpendicular from P to the radical axis and $t_1^2 - t_2^2 = K$. PN.AB then K =
1) 0 2) 1 3) 2 4) 1/2
- The locus of centres of all circles which touch the line $x = 2a$ and cut the circle $x^2 + y^2 = a^2$ orthogonally is
1) $y^2 + 4ax - 5a^2 = 0$ 2) $y^2 + 4ax + 5a^2 = 0$
3) $y^2 = 4ax + 5a^2$ 4) $y^2 = 4ax - 5a^2$

- If a circle passes through the point (a,b) and cuts the circle $x^2 + y^2 = k^2$ orthogonally then the locus of its centre is
1) $2ax + 2by = a^2 + b^2 + k^2$
2) $ax + by = a^2 + b^2 + k^2$
3) $x^2 + y^2 + 2ax + 2by + k^2 = 0$
4) $x^2 + y^2 - 2ax + 2by + k^2 = 0$
- The equation of the circle which passes through the origin has its centre on the line $x + y = 4$ and cuts orthogonally the circle $x^2 + y^2 - 4x + 2y + 4 = 0$
1) $x^2 + y^2 - 4x - 4y = 0$ 2) $x^2 + y^2 - 2x - 6y = 0$
3) $x^2 + y^2 - 6x - 2y = 0$ 4) $x^2 + y^2 + 4x - 12y = 0$
- The centre of circle cutting $x^2 + y^2 - 2x + 4y = 0$ orthogonally and passing through $(0,0), (2,0)$ is
1) $(3/2, 1)$ 2) $(1, 3/4)$ 3) $(1, -3/4)$ 4) $(-1, -3)$
- $x = 1$ is the radical axis of two circles which cuts each other orthogonally.
If $x^2 + y^2 - 8x + 4 = 0$ is the equation of one circle then the radius of the other circle is
1) 4 2) 2 3) 6 4) 3
- Let $n (\geq 3)$ circles having different radical axes and radical centres. The value of 'n' for which the number of radical axes is equal to the number of radical centres is
1) 3 2) 4 3) 5 4) 8
- A, B, C are the centres of three circles which cut each other orthogonally. The radical centre of the three circles is of the $\triangle ABC$
1) Incentre 2) Centroid
3) Ortho centre 4) Circum centre
- The radical centre of the three circles $x^2 + y^2 = 9$, $x^2 + y^2 - 2x - 2y - 5 = 0$ and $x^2 + y^2 + 4x + 6y - 19 = 0$ is
1) $(1, -1)$ 2) $(1, 2)$ 3) $(1, 1)$ 4) $(-1, -1)$

COMMON CHORD OF THE CIRCLES

- If $x - y + 1 = 0$ meets the circles $x^2 + y^2 + y - 1 = 0$ at A, B, then the equation of the circle with AB as diameter is
1) $2(x^2 + y^2) + 3x - y + 1 = 0$
2) $2(x^2 + y^2) + 3x - y + 2 = 0$
3) $2(x^2 + y^2) + 3x - y + 3 = 0$
4) $x^2 + y^2 + 3x - y + 1 = 0$

12. The line $2x + 3y = 1$ cuts the circle $x^2 + y^2 = 4$ in P and Q. Then the equation of the smaller circle passing through P and Q

- 1) $13(x^2 + y^2) - 4x - 6y + 50 = 0$
 2) $13(x^2 + y^2) - 6y - 50 = 0$
 3) $13(x^2 + y^2) - 4x - 6y - 50 = 0$
 4) $13(x^2 + y^2) - 4x - 50 = 0$

13. $(-2, 3)$ is the middle point of chord AB of the circle $x^2 + y^2 = 81$. The equation of the circle through the points A, B and $(0, 1)$ is

- 1) $x^2 + y^2 - 16x + 24y - 23 = 0$
 2) $x^2 + y^2 + 16x - 24y + 23 = 0$

- 3) $x^2 + y^2 - 2y + 1 = 0$ 4) $x^2 + y^2 - 16x - 24y = 0$

14. If the equation of the circle whose one diameter is the common chord of the circles $x^2 + y^2 + 4x + 2y - 4 = 0$ and $x^2 + y^2 - 2x - 4y - 4 = 0$ is $x^2 + y^2 + ax + by + c = 0$ then $a + b - c =$

- 1) 4 2) -4 3) -8 4) -2

15. The circle which cuts the circles

$$x^2 + y^2 + a_1x + b_1y + c = 0,$$

$$x^2 + y^2 + a_2x + b_2y + c = 0,$$

$$x^2 + y^2 + a_3x + b_3y + c = 0 \text{ orthogonally is}$$

- 1) $x^2 + y^2 + x - y = c$ 2) $x^2 + y^2 = 2c$
 3) $x^2 + y^2 = 3c$ 4) $x^2 + y^2 = c$

16. The equation of the circle which bisects the circumferences of the circles $x^2 + y^2 = 1$,

$$x^2 + y^2 + 2x = 3, \quad x^2 + y^2 + 2y = 3 \text{ is}$$

- 1) $x^2 + y^2 + 4x + 4y - 1 = 0$
 2) $x^2 + y^2 + 2x + 2y - 1 = 0$
 3) $x^2 + y^2 + 4x + 2y - 1 = 0$
 4) $x^2 + y^2 + 2x + 4y - 1 = 0$

17. The orthocentre of the triangle formed by the parametric points α, β and γ on the circle $x^2 + y^2 = a^2$ is

- 1) $(0, 0)$ 2) $\left(\sum \frac{a}{3} \cos \alpha, \sum \frac{a}{3} \sin \alpha\right)$
 3) $\left(\sum a \cos \alpha, \sum a \sin \alpha\right)$
 4) $\left(\sum a \cos \frac{\alpha}{2}, \sum a \sin \frac{\alpha}{2}\right)$

18. Three circles are such that each touch the other two externally. The common tangents are concurrent at 'p'. The length of the tangent to each circle is 'p'. The ratio of the product of their radii to sum of their radii is

- 1) $\frac{p}{2}$ 2) $\frac{p^2}{2}$ 3) p 4) p^2

19. A and B are two points on the circle $x^2 + y^2 = 1$. If the x co-ordination of A and B are the roots of the equation $x^2 + ax + b = 0$ and the y-coordinates of A and B are the roots of the equation $y^2 + by + a = 0$ then the equation of the line AB is

- 1) $ax + by = 0$ 2) $ax + by + 1 = 0$
 3) $bx + ay + a + b = 0$ 4) $ax + by + a + b + 1 = 0$

20. If the length of tangents from (a, b) to the circles $x^2 + y^2 - 4x - 5 = 0$ and $x^2 + y^2 + 6x - 2y + 6 = 0$ are equal then

- $10a - 2b =$
 1) 10 2) -11 3) 11 4) -10

LEVEL-II (H.W.) - KEY

- 1) 1 2) 3 3) 1 4) 1 5) 1 6) 2 7) 2
 8) 3 9) 3 10) 3 11) 1 12) 3 13) 2 14) 1
 15) 4 16) 1 17) 3 18) 4 19) 4 20) 2

LEVEL-I (H.W.) - HINTS

- $\cos \theta = \frac{d^2 - r_1^2 - r_2^2}{2r_1r_2}$ where $r_1 = r_2$ and $\theta = 120^\circ$
- Take A(a, 0), B(-a, 0) and P(x₁, y₁) and $r_1 = r_2 = a$
- Let S = 0 be req. circle and use given conditions
- Let the centre of the circle be (x₁, y₁)
- Let S = 0 be req. circle and use given conditions
- use orthogonality condition
- "C₂ = " C₃ $\Rightarrow n = 5$
- Intersection of R.A. of any two pairs of circle
- 11, 12. $S + \lambda L = 0$ has centre on L = 0
- Equation of AB is $S_1 = S_{11}$ and req circle is $S + \lambda(S_1 - S_{11}) = 0$
- Find the radical axis and then the circle
- Centre c = (0, 0), radius $r = \sqrt{c}$
- Circum centre (0, 0) $\Rightarrow H = 3G$
- $r_1 = p \cot \frac{A}{2}, r_2 = p \cot \frac{B}{2}, r_3 = p \cot \frac{C}{2}$
- $S = x^2 + y^2 - 1 = 0$
 $S' = x^2 + y^2 + ax + by + a + b = 0$ Find $S - S'$
- (a, b) lies on radical axis
